

GOLDFISH · CARASSIUS AURATUS

The bowl was always wrong

A complete, fact-checked printable manual: tank size, filtration, cycling, water chemistry, safe foods, health triage, and the routines that keep a goldfish thriving for its full lifespan.

INSIDE

- ✔ Tank size by variety, with filtration turnover math
- ✔ Safe foods charts: pellets, greens, protein and the never list
- ✔ Fishless and fish-in cycling, step by step
- ✔ Owner logs, pet-sitter sheet and a power-outage plan

48 pages

Beginner

Print or tablet

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Every page is written to be used, not just read once.

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How to use this package

Three ways to get value from it, depending on where you are right now.

UNITS USED THROUGHOUT

Every temperature and size is given twice. Temperatures appear in degrees Fahrenheit (°F) with degrees Celsius (°C) beside them. Lengths appear in inches (in) and feet (ft) with centimeters (cm) beside them, and the smallest in millimeters (mm). Weight is in pounds (lb). Tank volume appears in gallons (gal) with liters (L) beside it, small doses in milliliters (ml), and filter flow in gallons per hour (gph). Water test results are written in ppm (parts per million) or mg/L (milligrams per liter), two ways of writing the same strength of a dissolved substance in water. Prices are in United States dollars.

If you haven't set up yet

Read pages 6 to 13 in order before buying anything. The tank and the filter shape the next decade, and both are cheaper to get right the first time. Buy the tank before the fish, and plan on four to six weeks of fishless cycling (growing the bacteria that process waste) before a goldfish goes in. Then use the Budget & Shopping List on page 35 to buy once.

If your goldfish is already set up

Audit what you have against the Setup Checklist on page 33, and go to the Red Flags page (27) if anything seems off. Test the water first. Most goldfish problems that look like disease are a water reading.

If you already have a fish in an uncycled tank

This is the most common situation, and it is fixable. Go to the fish-in cycle on page 10. Do not tear the tank down or start dosing treatments: daily testing and water changes carry you through it.

Print these first

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| ☐ Page 33: Setup Checklist & Targets | ☐ Page 40: Pet-Sitter Sheet |
| ☐ Page 34: Emergency & Quick Targets Card | ☐ Page 41: Owner Log |
| ☐ Page 37: Symptom Quick Reference | ☐ Page 42: Equipment & Vet Log |
| ☐ Page 38: Daily, Weekly & Seasonal Routine | ☐ Page 43: Enrichment Checklist & Log |

WHY THIS PACKAGE EXISTS

The bowl, the line that a goldfish grows to fit its tank, and the three-second memory are still repeated, and all three are wrong. This package replaces them with numbers you can test against, drawn from the veterinary and university sources on page 46.

NOT A SUBSTITUTE FOR

A veterinarian who treats fish. This is a husbandry reference for day-to-day care, and anything urgent is a vet call. Most small-animal clinics do not see fish, so find a fish vet before you need one.

Quick profile

Everything at a glance before you read the detail.

Scientific name	<i>Carassius auratus</i> , bred from the Prussian carp
Adult size	Fancy varieties 6 to 8 in (15 to 20 cm); shubunkin 10 to 12 in (25 to 30 cm); common and comet 10 to 14+ in (25 to 36+ cm). Pond fish have passed 18 in (46 cm)
Captive lifespan	10 to 15 years in a well-kept tank, 20 to 30+ in a spacious pond. The record is 43 years
Origin	Bred in China over a thousand years ago from a wild carp relative
Diet type	Omnivore: a sinking goldfish pellet as the staple, with vegetables and protein foods alongside (page 14)
Tank minimum (adult)	20 gal (76 L) for one fancy, plus 10 gal (38 L) per extra fish; 40 gal (151 L) for a shubunkin; 55 gal (208 L) for one common or comet, and a pond is better
Water temperature	65 to 75°F (18 to 24°C). A coldwater fish: no heater unless the room falls below that
Water chemistry	Ammonia 0, nitrite 0, nitrate under 20 ppm, pH 6.5 to 7.5 (page 11)
Filtration minimum	The tank's volume turned over 10 times an hour or more (page 8)
Sexual maturity	About one year, with the prime years around the third (page 24)
Handling difficulty	Not a hands-on pet. Container transfers, a soft rubber net, never bare hands (page 19)
Social needs	Social: generally does better with its own kind than alone
Conservation status	Domesticated and not evaluated; the wild ancestor is Least Concern, the lowest-risk grade on the global list of threatened species

Cost overview

Initial setup

\$105 to \$270 for the equipment, plus under \$10 for the fish itself, more for a fancy variety. Every line is on page 35.

Monthly ongoing

\$10 to \$30 for pellets, occasional gel or frozen food, and periodic water treatments.

THE CHEAPEST ANIMAL IN THE STORE, ONE OF THE COSTLIEST FISH TO HOUSE

The fish costs next to nothing. The money is in the tank and the filter, and the expensive mistake is buying the cheap bowl first and the correct tank six months later, alongside a sick fish.

Fun facts worth knowing

- Goldfish form associative memories that last months and can be trained to push levers for food and navigate mazes, which makes the three-second memory line backwards.
- A goldfish has no true stomach, so food passes along the gut: small portions suit it, and one large meal does not.
- Goldfish and koi can be crossed, mostly by artificial spawning, and the first-generation hybrid males examined were sterile.

The goldfish, varieties, size & lifespan

What you are buying, how big it gets, and how long it stays.

One species, a thousand years of breeding

Every pet goldfish is *Carassius auratus*, bred in China from a wild carp relative for color mutations noted as early as the Jin dynasty (266 to 420). Moving the fish indoors in the Ming dynasty let breeders keep fancy-tailed mutations too fragile for an outdoor pond. The result is dozens of varieties, from the streamlined common and comet to the round-bodied oranda and ranchu.

Which goldfish do you actually have

TYPE	ADULT SIZE	MINIMUM FOR ONE	NOTES
Fancy: oranda, ranchu and other round-bodied varieties	6 to 8 in (15 to 20 cm)	20 gal (76 L), plus 10 gal (38 L) each	Round-bodied. Swim bladder trouble is most common in this group, and delicate fins need smooth surroundings. Some varieties, telescope eye among them, are bred with prominent eyes.
Shubunkin	10 to 12 in (25 to 30 cm)	40 gal (151 L)	A larger tank of 55 gal (208 L) or more, or a pond, is the better setup.
Common and comet	10 to 14+ in (25 to 36+ cm)	55 gal (208 L)	The single-tail fish sold as feeders. A feeder is the same species as a pet goldfish, and it is better suited to a pond than an indoor tank in the long run.

Size: it does not fit the tank

Goldfish do not stay small to match their container. A small tank stunts them: poor water, chronic ammonia and cramped space suppress outward growth while the organs and skeleton keep developing, linked to shortened lives and organ failure. In outdoor ponds with excellent water, goldfish regularly reach 12 in (30 cm) or more. The small goldfish most people picture is a stunted one.

Lifespan

Ten to 15 years is normal in a well-kept tank and 20 to 30 or more in a spacious pond. The oldest on record, a male named Tish, lived from 1956 to 1999, 43 years. That makes a goldfish a decade-long commitment.

Company

Goldfish are social and generally do better with their own kind than alone. Keep fancy with fancy and single-tail with single-tail, so that body shape and swim speed match and competition at feeding time stays low. A tropical tank is the wrong home for it: a goldfish needs cool water, and guppies and anything slower at the food are the pairings that fail.

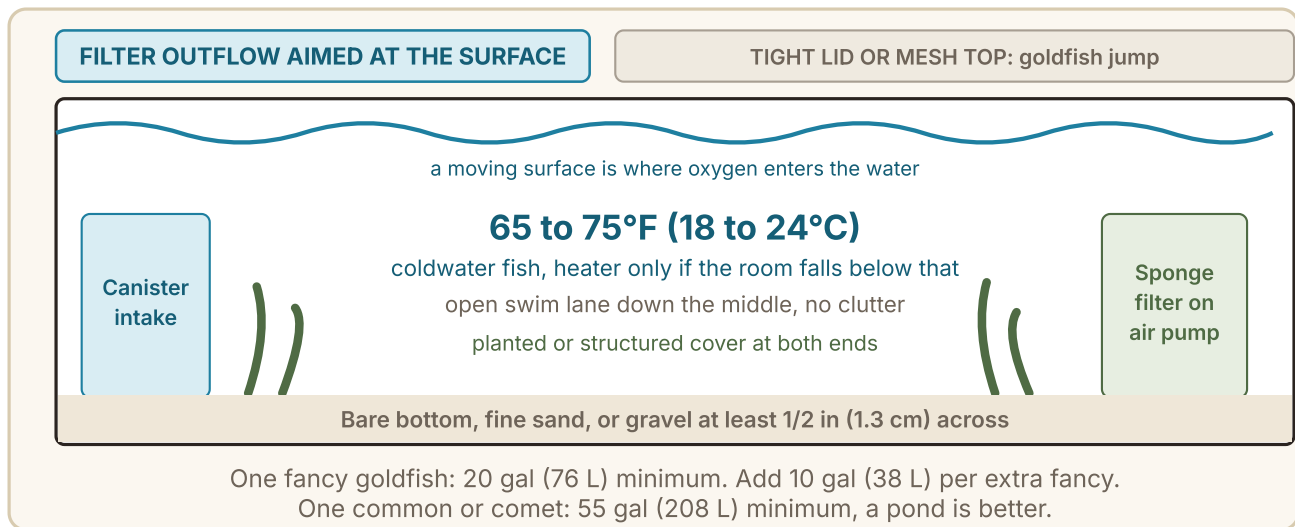
GOLDFISH OR KOI?

They are different species. A koi has two pairs of whisker-like barbels at the corners of its mouth, and a goldfish has none. A goldfish can carry koi herpesvirus without showing signs and pass it to koi, so a goldfish added to a koi pond is quarantined like a new koi.

Tank size, type & the bowl myth

A single fancy goldfish needs at least 20 gal (76 L), with 10 gal (38 L) added for each fish after that. A shubunkin needs 40 gal (151 L). A common or comet needs at least 55 gal (208 L) for one fish and belongs in an outdoor pond in the long run. Long and wide beats tall, because goldfish are horizontal swimmers. A 20-gallon long suits a fancy: it is sized for a 30 in (76 cm) length and a 12 in (30 cm) turning radius.

A bowl-kept goldfish often survives months or a couple of years, which is long enough for the setup to look like it works while the fish slowly declines. Without a filter, ammonia builds up within a day or two even in a large bowl.



NEVER

Keep a goldfish in a bowl, a vase or a self-cleaning kit with a plant growing from the top. A handful of plant roots cannot process the ammonia a goldfish produces, and none of those containers hold enough water or filtration. The marketing changed; the underlying problem did not.

Two fish, and how to tell the tank is too small

- Two fancy goldfish need at least 30 gal (114 L) with strong filtration. Two commons or comets need a pond or a tank well over 60 gal (227 L).
- The test is the water, not the fish. If ammonia or nitrite ever reads above zero between water changes, or you are changing water more than twice a week just to keep the fish alive, the tank is too small for the waste it carries.
- A few days in a bowl in an emergency, such as a move, is not housing. Even then it needs partial water changes every day.

Water temperature, heater & lid

A coldwater fish, so the equipment list is shorter than for a tropical tank.

The target

Keep the water at 65 to 75°F (18 to 24°C). At normal room temperature a goldfish needs no heater, and a heater running too warm stresses it. A heater is only for a room that falls below 65°F. A thermometer in the tank tells you which case you are in.

Where the tank goes

Away from a sunny window and away from anything that radiates heat, because the water slowly follows the room. A tank placed well solves much of the summer problem before the water goes in.

Tolerant, but only after adjusting

A goldfish survives a huge range, but its limits move with the water it has been living in. The coldest temperature goldfish survived ranged from 0.3 to 12.6°C (33 to 55°F) and the hottest from 30.8 to 43.6°C (87 to 110°F), and the temperature they had been acclimated to explained about 90% of that difference. So the rule is slowness. Change the temperature by no more than 1°F (about 0.5°C) an hour, and match the water before moving a fish between a warm shop, a room, a tub or a pond.

STEADY BEATS PERFECT

A tank held steady at a slightly too warm temperature is likely a better outcome than the same tank swinging between too cold and too warm in a day. Rapid temperature changes sit on the list of conditions that come before disease.

Why warm water is the danger

Warm water holds less oxygen while a warmer fish uses more. Water at 90°F (32°C) holds about 7.4 mg/L of dissolved oxygen when full, and water at 45°F (7°C) about 11.9 mg/L. Goldfish warmed from 21 to 30°C (70 to 86°F) used steadily more oxygen, roughly tripling for every 10°C (18°F) of warming. That is why a hot tank can crash with nothing else visibly wrong. The steps for a heat wave are on page 25.

DISSOLVED OXYGEN	WHAT IT MEANS
Above 5 mg/L	Supports optimum health
2 to 4 mg/L	Distress, with gasping at the surface as the warning sign
Under 2 mg/L	Deaths follow

The lid

Secure the tank with a tightly fitting lid to prevent escape. A mesh top does the same job and lets heat and humid air out, which matters in summer.

Filtration: sizing the filter

A single goldfish produces a waste load, its bioload, comparable to several tropical fish of the same size combined. That one fact drives almost every filtration decision here, and it is why experienced keepers run two filters as standard practice rather than as backup equipment.

FILTRATION TARGET	NUMBER
Turnover to aim for	The tank's whole volume 10 times an hour or more. That is the heavy column for a heavily stocked, messy tank
Lightly stocked or planted tank	About 4 times an hour
Typical community tank	6 to 8 times an hour
On a 20 gal (76 L) tank	200 gph, about 760 liters an hour
On a 40 gal (151 L) tank	400 gph, about 1,500 liters an hour
On a 55 gal (208 L) tank	550 gph, about 2,080 liters an hour

The box number is a best case

The flow rating printed on a filter is measured in clean water, with no media loaded and no height for the pump to lift against. Your filter never runs that way. Media clogs between cleanings and every foot of lift costs flow. Treat the rating as a ceiling you will rarely reach, size from the multiplication above rather than the tank-size label, and buy toward the higher end.

Filter types for a goldfish tank

TYPE	BEST FOR	TRADE-OFF
Hang-on-back	Standard tanks, the easiest to maintain	Less room for media than a canister
Canister	Large or heavily stocked tanks that need serious turnover	Most expensive, most involved to open and clean
Sponge, on an air pump	Gentle low-flow filtering, fry, hospital tanks	Little chemical filtration, needs a separate air pump

Running two filters on one goldfish tank is normal. A sponge filter on an air pump can run beside the main unit, and a spare sponge kept cycling in the main filter can seed a bare quarantine tank.

AIM THE OUTFLOW AT THE SURFACE

Oxygen enters the water at the surface, and a moving surface does it fastest. Pointing the filter outflow so it breaks the surface adds oxygen at no cost, and it matters most in summer, when warm water holds less. A battery-powered air pump is the one item worth owning before an outage (page 39).

PICK THE FILTER BY THE ORDER OF THE WORK

Mechanical media catches solids first, biological media turns ammonia into nitrite and then nitrate, and chemical media polishes last. The biological stage is the one that keeps fish alive day to day.

Filter media, air pump & maintenance

The dirtiest-looking part is the forgiving one. The cleanest-looking part is the one that keeps the fish alive.

MEDIA	WHAT IT REMOVES	HOW TO MAINTAIN IT
Mechanical floss, foam, sponge	Uneaten food, waste particles, visible debris	Rinse freely, even weekly. It holds no colony worth protecting.
Biological ceramic rings, sponge	Ammonia and nitrite, through the bacteria living on its surface	Barely ever. Rinse gently in old tank water only, and never all at once.
Chemical activated carbon	Dissolved toxins, discoloration, odor, residual chlorine	Consumable: replace it on a schedule. Take it out before any medication and keep it out until the course ends.

NEVER

Rinse biological media under the tap. Tap water carries chlorine or chloramine to kill microbes, and the media is covered in the microbes you want to keep. A few seconds under the tap is a small version of the disinfection that kills a whole colony. Fill a bucket with water siphoned out at a water change, swish the media in that, and put it straight back.

Replace biological media in portions

Swapping every piece of biological media in one pass removes the surface the colony lives on and forces a partial recycle. Replace it in portions, staggered weeks apart, so the untouched media keeps the colony running while the new media fills in.

After a medication

Antimicrobial treatments tend to kill the filter's bacteria along with their target, and copper is harder on it still. Expect ammonia and nitrite to rise for weeks to months after a copper course, and keep testing long after the course ends.

When the filter stops

The colony needs a steady flow of oxygenated water, and it becomes inefficient once dissolved oxygen in the filter drops below about 2 ppm. After an outage, a filter that sat dead for hours holds stagnant water and decomposing waste. Unplug it deliberately rather than leaving it dead in the water, and rinse the media in old tank water, or at least open the housing and look, before switching it back on.

The air pump

An air pump runs a sponge filter and adds surface movement. It pays for itself in an outage and again in a heat wave, when extra aeration is the cheapest margin you can buy.

Cycling, with or without a fish

Cycling means growing two groups of bacteria: one converts ammonia, which is toxic, into nitrite, which is also toxic, and the second converts nitrite into nitrate, which is far less toxic and is what routine water changes remove. Until the colony exists, a fish in the tank is living in its own waste.

The fishless cycle, four to six weeks

STEP	WHAT TO DO
1. Dose	Add pure ammonia, with no added surfactants or dyes, to a target of 3 ppm, inside the usual 2 to 4 ppm range.
2. Test and redose	Test daily or every other day, and redose back up to 3 ppm once it drops.
3. Wait	Most cycles take four to six weeks in warm, stable water. Real tanks have taken from about three weeks to two months. Overdosing slows it, because high ammonia inhibits the nitrite-converting bacteria.
4. Confirm	Done when a full dose of ammonia reads zero within 24 hours, nitrite reads zero, and nitrate has begun to build. All three must be true before a fish goes in.

Speeding it up. Media, gravel or filter floss from an established, healthy tank is the most reliable way to seed a new one. Bottled bacteria products are inconsistent, so do not assume one replaces the weeks of dosing and testing. Cold water and an inconsistent dose push a cycle toward the slow end.

The fish-in cycle

Many goldfish are bought on the same trip as the tank. This is the recovery plan, not a reason to start over.

1. **Test daily** for ammonia and nitrite with a liquid reagent kit.
2. **Act at 0.1 ppm.** A reading of 0.1 ppm of either calls for a water change of at least a third of the tank. A reading clearly climbing toward toxic calls for 50 percent or more, done without hesitation: a large change that briefly interrupts the cycle is a far smaller risk than leaving a fish in rising ammonia. Most fish-in cycles are managed day to day between 0.1 and 0.25 ppm.
3. **Use an ammonia-binding conditioner.** It buys a margin between changes without killing the bacteria you are growing. It is a bridge, not a substitute for testing and changes.
4. **Leave the filter alone.** No tap rinsing, no replacing all the media at once.

HOW YOU KNOW IT IS FINISHED

Ammonia and nitrite hold at zero and nitrate builds. Nitrate is the proof, because it is the end product of both conversions. From then on, nitrate is what your weekly water changes remove (page 12).

IF A CYCLED TANK READS AMMONIA AGAIN

Work backward through the usual causes before blaming the fish: overdosed ammonia, a pH or KH (carbonate hardness, the buffer that holds pH steady) crash, untreated tap water, a medication, tap-rinsed media, or a cold drop. A change of 1°C (about 2°F) can cut the bacteria's work by around 30 percent.

Water targets & testing

Every number on this page is testable. None of it can be judged by looking at the tank.

READING	TARGET	ACT WHEN
Ammonia	0 ppm	Anything above 0 is an action today
Nitrite	0 ppm	Anything above 0 is an action today
Nitrate	Under 20 ppm	Past 40 ppm, the schedule has failed and the next change gets bigger
pH	6.5 to 7.5	Anywhere from 6.5 to 9.0 is inside the veterinary reference range. Steady matters more than the exact figure
Temperature	65 to 75°F (18 to 24°C)	Heating or cooling by more than 1°F (0.5°C) an hour
GH (general hardness: the dissolved calcium and magnesium)	61 to 180 mg/L	Moderately hard to hard water suits most goldfish
KH (carbonate hardness: the water's buffer against pH change)	Above 100 mg/L	At or below 50 mg/L the biofilter suffers
Dissolved oxygen	Above 5 mg/L	2 to 4 mg/L is distress, under 2 is deadly

Reading the numbers together

Ammonia and nitrite are the emergency pair, both toxic, both exactly zero in a cycled tank. Unionized ammonia above about 0.05 mg/L already damages gill tissue, and chronic exposure brings lethargy and poor appetite. Nitrate is the housekeeping number: most fish tolerate up to about 20 mg/L, and 40 is the ceiling below which most are considered safe. It rises between changes, so it measures whether your water-change schedule keeps up with your stocking and feeding.

KH is the buffer that holds pH still. When it is used up, a tank can crash in pH suddenly, and a bottle of pH-up or pH-down against low KH can overshoot badly. Reach for a plain water change, or crushed coral or aragonite for a slow safe lift in KH and GH, before a bottled adjuster. Goldfish have far more room than a tropical fish, which is why the crash surprises people after years of neglect.

USE A LIQUID REAGENT KIT

A kit that reads ammonia, nitrite, nitrate and pH, in liquid form, is the only reliable way to know the numbers rather than guess from how the fish look. A clear tank is not a safe tank: ammonia, nitrite and low oxygen are all invisible. Test first whenever a fish looks off (page 28).

CHLORAMINE DOES NOT EVAPORATE

Many water supplies use chloramine, chlorine bonded to ammonia, so standing a bucket overnight does not make the water safe. Use a conditioner labeled for both chlorine and chloramine, every time.

Water changes, cleaning & hygiene

A filter never removes nitrate. Only taking water out does.

How much, and how often

Change 25 to 50 percent of the water weekly on a stocked goldfish tank, because a goldfish is the heaviest waste producer for its size in the hobby. Test nitrate before each change and let the reading set the size.

NITRATE BEFORE THE CHANGE	WHAT IT MEANS	WHAT TO DO
Under 20 ppm	The schedule matches the waste	Keep the weekly change
20 to 40 ppm	Nitrate is winning slowly	Change 40 to 50 percent weekly, or add a midweek change, and feed less
Over 40 ppm	Past the safe line	A 50 percent change now with matched water, retest in two days, and find the cause

Small changes run against you: a 25 percent change on a tank that adds 20 ppm a week settles near 80 ppm, while 50 percent settles near 40.

The steps

1. Test first and write the numbers down. Leave the filter alone: you are not cleaning it today.
2. Siphon from the substrate up, since most of the waste settles into it, and most nitrate is made there.
3. Match three things before the water goes back: no chlorine or chloramine, a temperature within a couple of degrees of the tank, and roughly the same chemistry.
4. Dose the conditioner as the water goes in, not after, because chlorine damages gills within minutes.
5. Rinse any media you must touch in the bucket of removed water, never under the tap.

NEVER

Swap most of the water in one go on a tank that has gone months without a change. The water has drifted far from the tap, and a catch-up change of 75 percent has killed half a tank overnight. That tank gets small daily changes: 10 to 15 percent a day for a week, every other day in the second week, then 20 to 25 percent twice a week for two more weeks.

Hygiene

Tank water can carry *Mycobacterium marinum*, a bacterium that infects people through cuts. Wear gloves whenever your hands are in the fish, the water or the equipment, and wash your hands as soon as the gloves come off. Keep separate nets and siphons for the hospital tank. A hospital tank and its gear are disinfected and stored dry between uses (page 23).

Substrate, plants & decor

Substrate: bare, sand, or big gravel

Goldfish forage by sucking sand and gravel into the mouth and spitting out what is not food. Occasionally a piece lodges in the mouth, and a lodged stone can need sedation and forceps at a fish vet to remove it without tearing the lining. Swallowed gravel is worse, because a goldfish cannot digest it and it can block the gut. That leaves three safe floors.

FLOOR	WHY
Bare bottom	Removes the risk entirely and is the easiest to keep clean
Fine sand	Lets the fish sift the way it wants to without swallowing harmful amounts. Coarse sand supports more natural foraging than larger gravel
Gravel at least 1/2 in (1.3 cm) across	Safe only when every piece is too big for the mouth. Small pea gravel, the default in many starter kits, is the size that causes trouble

Plants

Goldfish eat and uproot most soft aquarium plants, so a planted goldfish tank is built from the tough ones. Anubias, especially the larger *Anubias barteri*, java fern and bolbitis fern have thick leaves and rhizomes that survive being bumped and chewed. All three grow attached to rock or driftwood instead of rooted in the substrate: tie or wedge them on and never bury the rhizome. Crinum calamistratum, a tough bulb plant that prefers cooler water, and marimo moss balls round out the list. Expect some damage whatever you pick.

Structure matters more than whether it is alive. Goldfish spend roughly 90 percent of their time in planted areas and about 10 percent in barren open space, with no strong preference for live over closely matched artificial plants.

Decor checklist

- Smooth hardscape only: fancy goldfish have delicate fins and some have prominent eyes, so sharp ornaments and rough rock are injury risks.
- Plants tied to wood or rock, not buried.
- An open swim lane running the length of the tank, with planted or structured cover at both ends.
- Everything secured so nothing can topple onto a fish.
- No tight plastic plants or fin-trapping clutter, which can tear delicate fancy fins.

ENRICHMENT STARTS HERE

Space, structure and a foraging-friendly floor are the biggest enrichment items a goldfish gets. The ranked list is on page 21.

Diet by age

Goldfish have no true stomach. Food passes along the gut instead of being held and released, which is why small portions suit them, a single large meal does not, and overfeeding shows up fast as a waste load the filter was not sized for.

AGE	FOOD	FEEDINGS	HOW MUCH
Fry, first couple of days	Nothing: still absorbing the yolk sac	None	Feeding now only fouls the water
Fry, free-swimming to about 1 month	Infusoria or baby brine shrimp nauplii (newly hatched shrimp)	3 to 6 a day	Small amounts
1 to about 4 months	Same foods	About 3 a day	Small amounts, fewer feedings than the first month
Juvenile	Sinking goldfish pellet, with blanched vegetables 2 to 3 times a week	2 to 3 small feedings a day	Higher caloric needs for growth
Adult	Sinking goldfish pellet, with blanched vegetables and protein foods 2 to 3 times a week	Once a day	A portion it finishes in under 2 minutes

The adult day's ration can be split into two smaller meals, which suits a fish with no stomach to hold a large one. The total stays the same. Adding meals without shrinking them is how most goldfish get overfed.

PORTION, NOT SCHEDULE

Feed only what is eaten in about 30 seconds to 2 minutes, and remove anything left. If the water clouds or nitrate climbs between changes, the fish is getting more than the tank can process. What to do about a high nitrate reading is on page 12.

Fasting is safe in a healthy adult

A healthy adult in a mature, well-kept tank can go 3 to 4 days without food with no real concern. That is why a short trip needs no automatic feeder. The limits and the warning signs are on page 18.

NO SUPPLEMENTS, AND THAT IS NORMAL

Unlike a reptile or an invertebrate, a goldfish needs no calcium dusting and no gut-loaded feeders. Nutritional balance comes from variety in the diet itself: a staple pellet, vegetables a couple of times a week, and protein foods in moderation.

Pellets, gel, flake & presentation

All three sit on the same shelf. They are not interchangeable.

FORMAT	VERDICT	NOTES
Sinking goldfish pellet	STAPLE	Cooked, more nutritionally uniform per particle than flake, formulated for a coldwater omnivore. The simplest, best-established staple.
Gel food	REASONABLE	Holds together in water far better than flake, so less protein and fat is lost into the tank. Good for picky eaters and for working medication into food. It has not replaced the sinking pellet as the default.
Goldfish flake	OCCASIONAL	Works for smaller or younger fish. It loses a share of certain vitamins within about 30 seconds of hitting the water and is easy to overfeed.
Tropical fish flake	AVOID AS STAPLE	Not toxic, but the nutrient profile is built for a different fish.

Why sinking matters

Fancy goldfish have a swim bladder connected directly to the digestive tract. Air swallowed while chasing floating food at the surface can end up in it and contribute to buoyancy problems. A sinking or neutrally buoyant diet cuts that air intake, which is why sinking pellets are generally the better call for round-bodied varieties (page 31). Sinking pellets also encourage the same natural foraging as a sand floor.

Keeping food fresh

Ascorbic acid, a vitamin goldfish need for normal growth, breaks down quickly once a container is open. Buy food in small amounts, ideally about once a month, store it cool and dry, and discard it after roughly three months whatever is left.

Feeding a pond goldfish by temperature

Outdoor fish scale their feeding to the water temperature, because their metabolism does.

WATER TEMPERATURE	WHAT TO FEED
Above 50°F (10°C)	Feed normally, scaled to how quickly it is taken
39 to 50°F (4 to 10°C)	A reduced, more digestible wheat-germ formula, fed less often
Below 39°F (4°C)	Stop feeding. The fish enters a low-metabolism dormancy and uneaten food only fouls the water

This applies to outdoor ponds. An indoor tank at room temperature stays in the top row all year, and a goldfish that turns sluggish indoors has a water or health problem, not a season.

Vegetables, protein foods & treats

A staple every meal, supplements a couple of times a week, treats rarely.

TIER	FOODS	WHY
STAPLE every meal	A sinking pellet formulated for goldfish	Complete, balanced nutrition for a coldwater omnivore
2 TO 3 TIMES A WEEK	Blanched peas, spinach, kale, lettuce, zucchini, cucumber; bloodworms, brine shrimp, daphnia; algae wafers	Adds variety and fiber without crowding out the pellet base
RARE TREAT	A small bit of orange, watermelon or squash	Fine now and then, not complete or balanced on its own

Blanching vegetables

Leafy and fibrous vegetables are good goldfish food, but not straight from the fridge. A goldfish has no teeth built for grinding raw fiber, so it mouths a chunk and spits it back. A minute or two in boiling water, then cooled, softens the structure enough to break apart. Cut the pieces small and offer only what will be eaten in a few minutes. Pull out whatever is left, because uneaten vegetable matter breaks down and fouls the water quickly.

Peas

Shelled peas, mashed or whole, are the genuinely useful short-term fix for constipation (page 31). They should not become a long-term staple.

Protein foods

Bloodworms, brine shrimp and daphnia are high-protein treats, fed in moderation as variety rather than as a second staple. Baby brine shrimp are the first food for free-swimming fry.

IF A FOOD IS NOT ON THIS PAGE

Treat that as a reason to skip it, not a reason to try it. The lists here are the foods with a reason behind them.

The never-feed list

Human food is where most well-meant feeding goes wrong.

ITEM	VERDICT	NOTES
Bread and other refined human starches	NEVER	Not a goldfish food in any amount.
Raw meat	NEVER	Not a goldfish food in any amount.
Cheese	NEVER	Not a goldfish food in any amount.
Onions	NEVER	Not a goldfish food in any amount.
Tropical fish flake as the main food	AVOID	Not toxic, but built for a different fish. Skip it as a primary food.
Citrus as a routine food	RARE ONLY	A small bit as a rare treat is the limit.

THE NEVER-FEED LIST, IN ONE PLACE

Bread and other refined human starches, raw meat, cheese and onions. A food that is not on the safe list is a food to skip.

Avocado and chocolate

Keep both away from any pet. The toxicity behind them is documented in birds and mammals rather than in fish, so the reason to avoid them here is that neither is a goldfish food, not a measured fish poison.

Whatever is left, pull it out

Uneaten food of any kind is one of the more common drivers of the ammonia spikes that stall a tank. Overfeeding with vegetables counts too. Remove leftovers after a couple of minutes and check the water if the glass clouds.

When a goldfish stops eating

Appetite loss is a symptom. Start with the water.

The usual reasons, in order

1. **Water quality.** Ammonia, nitrite or nitrate spikes, especially in a newer tank still cycling, are the most common cause.
2. **Overfeeding, constipation or swim bladder trouble.** Excess or low-fiber food causes gas buildup that presses on the swim bladder, so appetite loss comes with abnormal floating or sinking.
3. **Stress.** A new tank, incompatible tankmates, bullying or a tank that is too small. Appetite loss is often the first visible sign.
4. **Disease.** Ich, internal or external parasites and fungal infections often cut appetite before anything else shows.
5. **Cold.** Below roughly 50°F (10°C) a pond goldfish slows into a natural low-metabolism dormancy and eats little to nothing until spring. That is normal for a pond fish, not illness.
6. **Breeding season.** As temperatures rise in late spring, goldfish getting ready to spawn, especially chasing males, may go off food for a while. It is hormone-driven and normal.
7. **Diet monotony.** A repetitive diet, or too many treats crowding out the pellets, can reduce interest.

How long is too long

TIME WITHOUT FOOD	WHAT IT MEANS
Up to 3 to 4 days	Fine for a healthy adult in an established tank, with no special arrangement
Beyond about 5 days	Effects on health may start to show
A week to 10 days	Arrange a feeder
1 to 2 weeks	The usual survival limit in a cold-water metabolism. It is an upper bound, never a target

THE TRIGGER TO GET HELP

24 to 48 hours of refusing food **combined with** any of: pronounced lethargy, scales standing out in a pinecone pattern, clamped fins, floating sideways or sinking, gasping at the surface, visible wounds or white spots, cotton-like growths, or a gaunt, thinning body. That combination, not the fasting alone, is what calls for a fish vet. Test the water first (page 27).

A GOLDFISH OFF FOOD IN A TOO-SMALL TANK

It is telling you about the tank. Check the volume and the filter before anything else.

Handling: net and cup, never hands

A goldfish is not a pet you interact with by touch, and that is how the species works.

Why the slime coat matters

A goldfish's skin is covered in a slime coat that is its first barrier against parasites, bacteria and infection, and it protects the scales. Bare hands, especially dry ones, wipe it off on contact. Contact like that decreases the fish's chemical protection, disrupts the regulation of salts and water balance in its body, and removes the physical barrier against invaders. The coat can take 24 to 48 hours to regenerate, and the fish is more open to infection for that whole window. Some of the fungal and bacterial problems goldfish pick up trace back to exactly this opening.

The right way to move a goldfish

1. **Guide it into a container of tank water** instead of lifting it out. It is the gentlest option and avoids the slime-coat problem almost entirely.
2. **If you need a net**, wet it first and choose soft rubber over knotted nylon, which catches on fins and scales. Use the net as a guide: steer the fish toward the container, then lift the container.
3. **If bare hands are unavoidable** in an emergency, wet them first and keep well clear of the gills and eyes, both especially sensitive.
4. **Keep the time out of water as short as you can**, whichever method you use.

WHAT HANDLING LOOKS LIKE DAY TO DAY

Container transfers for water changes, gentle net use when moving a fish to a new tank, and otherwise leaving it in its own water. The relationship is built through water quality, feeding and observation, which is plenty for a fish that can live well over a decade.

Carrying a goldfish any distance

A trip to a vet or a new home is a matter of oxygen and temperature in the travel container. The bagging steps, the air gap and the battery air pump are on page 39.

YOUR OWN HANDS AND THE TANK

Tank water can carry *Mycobacterium marinum*, which infects people through cuts. Wear gloves for contact with the fish, the water or the equipment, and wash your hands as soon as the gloves come off.

Normal behavior & body language

Behavior changes show up before anything visible does. Learn the normal first.

What a goldfish is like

Goldfish are social and generally do better with their own kind. They form associative memories that last months, they can be trained to navigate mazes and push levers for food, and they appear to learn the routines of the person who feeds them. That is not a three-second memory.

BEHAVIOR	VERDICT	WHAT IT USUALLY MEANS
Sifting sand or gravel and spitting some out	NORMAL	Foraging. This is what the substrate is for.
Chasing and nudging in late spring in a mixed group	NORMAL	Spawning behavior, hormone-driven. Males chase, and some go off food for a while.
Little activity in cold pond water	NORMAL	Seasonal dormancy in a pond. Indoors, a sluggish goldfish has a water or health problem.
Rubbing against decor, gravel or glass (flashing)	LOOK CLOSER	An irritant on the skin or gills, commonly an external parasite, or simple water chemistry irritation. Test the water first.
Gulping at the surface (piping)	ACT	Usually low oxygen, not infection. Test, add aeration and surface movement.
Hanging in the current from the filter or air stone	LOOK CLOSER	Often seeking the most oxygen-rich water. Test the water.
Fins clamped tight to the body, with lethargy	ACT	One of the more universal signs of stress or illness.
Hiding, isolating, or resting somewhere unusual	LOOK CLOSER	Worth a closer look the same day, not tomorrow.
Off food, with any other sign	ACT	See the appetite page and the red flags.

CURRENT

A mild current from adequate filtration can encourage natural swimming. Avoid exhausting flow, especially for heavy-bodied fancy varieties that are not built for strong currents.

COLOR

A fish looking washed out, paler or duller than its normal color is a general illness indicator that often shows before anything more specific, which is what makes it useful as a keep-watching sign.

Enrichment & common mistakes

Mistakes that cause the most harm

- **Adding fish to an uncycled tank.** There is no shortcut that beats testing until ammonia and nitrite both read zero.
- **Under-filtering because the tank looks clean.** Ammonia and nitrite are invisible, and a goldfish produces far more of them than its size suggests.
- **Overfeeding.** Adding meals without shrinking them is how most goldfish get overfed, and every uneaten pellet becomes ammonia.
- **Buying the fish before the tank.** The fish costs a few dollars and the setup costs a few hundred. Doing it in that order is what produces the bowl.
- **Cleaning too thoroughly.** Rinsing media under the tap or replacing all of it strips out the bacteria colony.
- **Mixing goldfish with tropical fish.** A goldfish needs cool water and a tropical community needs a heater, and goldfish will go after the fins of slow, long-finned fish such as a betta.

Enrichment, by priority

Enrichment comes after correct volume, filtration and a fully cycled tank, never instead of them. Decor cannot rescue a bowl or chronically poor water. In this order:

1. SWIMMING VOLUME

A long, wide footprint with a deliberately open lane down the middle matters as much as any structure.

2. STRUCTURE AND COVER

Fish spent about 90 percent of their time in planted areas and 10 percent in barren space, live or artificial.

3. FORAGING FLOOR

Sand supported more natural foraging than larger gravel, and goldfish kept foraging even when not hungry.

4. SINKING FOOD

Sinking pellets as the staple encourage the same foraging, which also helps cut air gulping in fancy varieties.

5. SAME-TYPE COMPANY

Fancy with fancy, single-tail with single-tail, only if the volume and filter carry the combined waste.

6. OCCASIONAL GREENS

Blanched vegetables add variety, within the same daily portion.

NOT ENRICHMENT

Overstocking called company, plants used as an excuse for weak filtration, and extra food called a treat are the same mistake wearing three different names. Sharp decor and fin-ripping plastic plants cause real injury. A bare, sterile box with no structure and no foraging outlet is under-enriched even with perfect water.

ENRICHMENT IS NOT EXTRA FOOD

Every foraging idea here uses the existing daily ration, delivered differently.

Choosing a healthy goldfish & bringing it home

Feeder or not

A feeder goldfish is the same species as a pet goldfish. The label describes what the fish was sold for, food for a larger animal, and a fish held on those terms is bought in bulk and kept crowded. The fish can still make a pet. The tank it comes from decides how risky it is, so judge the tank before the fish.

- Walk away from any tank holding a dead or visibly sick fish, whatever the price.
- A healthy goldfish has clear eyes, a normal body condition, intact skin and fins, no wounds, and breathes at the same pace as its tankmates. Fast gill movement next to calm neighbors is a warning.
- Visit the shop a few times before buying, to see whether its animals look healthy in general or only on a good day.

A FISH THAT PASSES EVERY CHECK CAN STILL CARRY SOMETHING

The water a goldfish comes home in can carry fish pathogens, among them *Aeromonas* and *Flavobacterium columnare*. Heavy intestinal infection with the parasite *Goussia* has been tied to high death rates in comet goldfish, with the fish lethargic, lying on their sides and passing pale feces.

Bringing it home

1. **Float the sealed bag** on the tank surface for 20 to 30 minutes so the water inside comes to the tank's temperature.
2. **Net the fish across**, or pour the bag through a net held over a bucket, and release only the fish. Bag water never goes into the tank, because it carries the shop's ammonia and whatever the shop's tanks were carrying.
3. **Go slow when the gap is big**. When the bag and the tank differ by more than 10°F (about 6°C), acclimate slowly with an airline running into the bag and tank water added gradually, and past a gap of about 18°F (10°C) stretch it over two hours or more. The same applies when the shop's water and yours sit at very different pH or hardness.

TEMPERATURE FIRST

Goldfish are often sold from warm shop tanks and taken home to an unheated tank or a garden pond. A fish that would shrug off cold water in December can be stressed by the same drop when it comes all at once. Match the temperature first, then keep the newcomer apart for its first weeks.

Quarantine @ the hospital tank

A fish that looks fine on arrival is the normal state of a fish whose symptoms have not shown yet. The value of a quarantine tank is that if something surfaces, it surfaces in an empty tank you can treat without risking every other fish you own.

PIECE	WHAT IT NEEDS
How long	At least 30 days. Thirty is the floor, not the target, and 30 to 60 days is the range for a fish you especially do not want to lose or to introduce something from.
The tank	A bare 10 to 20 gal (38 to 76 L) tank sized to what you quarantine: no substrate, wood or rock to hide a parasite or its eggs.
Filter	A sponge filter and a small air pump, seeded from a mature filter, either from a spare sponge kept cycling in the main filter or borrowed for a few days, so it works on day one.
Heater	None at room temperature. Only if the room falls below 65°F or a vet's treatment calls for it.
Gear	Its own nets and siphon hoses that never touch another tank. Disinfect the tank and gear between uses and store them dry.

What to watch for

Feed normally, do routine partial water changes, and check daily for anything not there on arrival: clamped fins, unusual spots or patches, labored or rapid gill movement, flashing, appetite loss, cloudy eyes, or a fish hanging listlessly. A fish that arrives thin or stressed may take a few days to show its real baseline, so do not read the first day or two too literally. The hospital filter is small and new, so test ammonia and nitrite partway through.

Treat there, not in the display

- Several medications kill the display's filter bacteria along with the pathogen, which stacks a poisoning risk on the illness.
- Copper is extremely toxic to invertebrates and plants, which must come out before it is used.
- **Pull the carbon first** and keep it out until the course ends, because carbon removes medication from the water.
- **Turn up the aeration**, since several treatments cut the oxygen available and warm water holds less.
- **Dose to the label or a vet**, never to memory.

AQUARIUM SALT

Salt is for scaled fish, which a goldfish is. The starting dose for ich is about one tablespoon of aquarium salt per 5 gallons, added gradually. There is no fixed number of days: continue until the infection is gone, then remove it with water changes. Salt does not evaporate, so top off with plain conditioned water. Ich is on page 28.

FIND THE VET DURING QUARANTINE

Most small-animal clinics do not see fish, so use the directory on the red flags page (27) to find one now, while nothing is wrong.

Sexing, spawning & fry

Goldfish are mature at about one year and in their prime in the third.

SIGN	WHAT TO LOOK FOR
Male, ready to spawn	Tubercles: small white pimples about the size of a pinhead, on the pectoral fins and the gill covers.
Female, ready to spawn	Heavy at the rear, with a swelling on the left side of the body. She can lay roughly 500 to 1,000 eggs in one spawn.
Timing	Goldfish usually spawn in warmer water, in the morning. In a mixed tank the chasing starts in late spring.

TUBERCLES ARE NOT ICH

Tubercles sit only on the pectoral fins and gill covers. The white spots of ich are small and spread evenly over the body and fins, and they come with flashing, clamped fins and lethargy (page 28).

Spring spawning behavior

As temperatures rise in late spring, mature goldfish in a mixed group may spawn. Males chase persistently, and some of the fish go off food for a while. That is hormone-driven and normal. Other fish prey on the eggs, which is the usual outcome unless you deliberately set up to raise fry.

Fry in brief

Do not feed for the first couple of days after hatching, because the fry are still absorbing the yolk sac. Once free-swimming, offer infusoria or baby brine shrimp nauplii, 3 to 6 times a day for the first month. The full schedule is on page 14.

A SWOLLEN BELLY IS NOT ALWAYS SPAWNING

A belly that is swollen with scales standing out in a pinecone pattern is not a fish ready to spawn. It is dropsy, and it is an emergency (page 29). Raising a full spawn of fry is beyond this package.

Heat waves, pond winters & moving outdoors

A hot tank, step by step

Warm water holds less oxygen while a goldfish uses more, so heat is a double bind. These methods work, ranked from the best:

METHOD	WHERE IT FITS
Room air conditioning	The single most effective option: the tank follows the room down.
A fan across the surface	The best cheap active method. It cools by evaporation and needs an open top or a mesh lid instead of glass.
Lights off, lid cracked open	Free and immediate. Lighting runs warm and a sealed lid traps warm, humid air.
Lower canister flow	A modest assist only. A canister motor adds heat, but do not cut below the turnover the bioload needs.
Shade and insulation	Prevents a slow daytime climb. Does little to cool water that is already hot.
Frozen bottles	Emergency stopgap for a single hot afternoon. They chill hard and then do nothing, which swings the temperature.
A chiller	The real fix when the problem recurs or the room runs warm all year.

Run extra aeration the whole time. Fish gulping at the surface of a pond are showing oxygen depletion, and more aeration is the best response. Do not change the temperature faster than 1°F (0.5°C) an hour.

Pond winters

Goldfish and koi are cold-hardy and can overwinter in an outdoor pond in most temperate climates, slowing their metabolism and settling near the bottom as the water cools. Feed by temperature, with the table on page 15, and stop below 39°F (4°C).

Moving between a tank and a pond

The same tolerance rule applies. A fish acclimated to a warm room is not ready for spring pond water, even though goldfish overwinter outdoors. Move it when the pond and the tank are close in temperature, and float it exactly as you would a new arrival. A pond goldfish that wintered outside has its limits set for cold water, and dropping it straight into a heated indoor tank is the same shock in reverse.

SPRING

Expect chasing in a mature group and a dip in appetite as the water warms.

Never release a goldfish: the law

Keeping a goldfish is legal in the states below. Letting one go is not.

STATE	STATUS	WHAT IT MEANS
Minnesota	Regulated invasive species	Legal to possess, sell, buy and transport, and illegal to release into the environment.
New York	Regulated invasive species (6 NYCRR, the New York Codes, Rules and Regulations, Part 575)	Legal to possess, sell, buy, propagate and transport. It may only not be knowingly introduced into a free-living state, and a shop selling it must label it as a regulated species.

Minnesota and New York rules as of October 2026.

THIS IS NOT A SURVEY OF ALL FIFTY STATES

These two states are the ones this package has a source for. Goldfish rules in your state are set by its wildlife agency, so check there. Nothing on this page is legal advice.

Why release is a problem

Goldfish released into lakes, ponds and rivers are an invasive species. In the wild they do not stay small: wild populations can reach close to 2 ft (60 cm) long. Common and comet goldfish are the ones that outgrow a tank, which is why the 55 gal (208 L) minimum and a pond plan matter before you buy.

If you cannot keep it

- Ask the retailer about a return.
- Find a surrender event to rehome the fish.
- Give or trade the fish with other hobbyists or a local society.
- Ask a veterinarian or a pet retailer about humane disposal. The method for a fish that will not recover is on page 32.

Red flags & finding a vet

Goldfish are hardier than their reputation when kept right. Most trouble starts in the water.

TEST THE WATER FIRST, THEN ACT, IF YOU SEE

Gasping at the surface or laboring gills. Scales standing out in a pinecone pattern. Floating, sinking or swimming at a persistent angle. Gray-white cottony growths. Open sores or red inflamed patches. White spots like grains of salt. Clamped fins with lethargy. A protruding eye. Refusing food for 24 to 48 hours alongside any other sign here.

Why the water comes first

Ammonia and nitrite are invisible, and a very large share of sudden sick-fish cases trace back to them, or to low oxygen, rather than to disease. Unionized ammonia above about 0.05 mg/L already damages gills. A fish in a tank less than a month old that looks sick points to the nitrogen cycle first. Ich, fin rot, fungus and ulcers all take hold in fish already stressed by poor water, crowding or a temperature swing, which is why treating the symptom without fixing the water buys little. Medicating before you rule the water out adds a stressor to a fish that already has one, and some medications also kill the filter's bacteria.

When it is a vet, not a home fix

- Illness spreading across more than one fish.
- An open ulcer, significant bloating or a badly protruding eye.
- Fin erosion or a white-spot outbreak advancing day by day despite clean water tests.
- A fish listing to one side, or resting on the bottom unresponsive.
- Persistent loss of buoyancy, severe fin damage, ongoing lethargy or appetite refusal, or a treatment that is not working.
- Any time the honest answer to "what caused this" is "I don't know, and the water tested clean". A vet can look at a skin or fecal sample under a microscope, which is the only reliable way to name a nonspecific sign.

When you call, have your recent water test numbers, the tank volume and filter, how long the signs have been going on, and anything you have already added to the tank.

Finding a vet who treats fish

Most small-animal clinics do not see fish, and a general search will not say which ones do. The American Association of Fish Veterinarians keeps a Find a Fish Vet map, which covers North America only. If it shows no one within reach, there are three next moves:

- Ask a local vet, ideally one who already sees your other pets, whether they would take the case with guidance from a fish vet. The association will connect the two.
- Contact the association directly. It will email its members to look for someone near you.
- Ask an aquarium shop or a local aquarium society, which usually know who in the area treats fish.

Ammonia poisoning & ich

Ammonia and nitrite poisoning

Cause: an uncycled tank, a tank overstocked for its filtration, overfeeding, or a filter crash after media was rinsed in tap water. It is entirely preventable. **Signs:** gill damage, lethargy, poor appetite and gasping, and in acute poisoning spinning and disorientation that can pass for an exotic disease. **Diagnosis:** a test kit, not the fish's appearance. Any reading above 0 for ammonia or nitrite is the diagnosis. **Response:** a water change of at least a third at 0.1 ppm, and 50 percent or more once a reading is clearly climbing toward toxic, with dechlorinated, temperature-matched water. Add surface movement. Do not deep-clean the filter, which removes the little colony you have.

ONE PRODUCT WORTH HAVING ON THE SHELF

An ammonia-binding water conditioner buys a margin between changes while the bacteria catch up. It is a bridge, not a substitute for a water change or for enough filtration.

Ich, or white spot disease

Cause: the parasite *Ichthyophthirius multifiliis*, almost always introduced on a new fish, plant or unquarantined decor, and taking hold in stressed fish. **Signs:** tiny white spots like grains of salt, evenly spread over the body and fins, with flashing, clamped fins, lethargy, dark coloration and rapid breathing if the gills are affected.

Why timing decides everything. The white spot is a stage armored by the fish's own skin. When it matures it drops to the bottom and forms a cyst, inside which it divides into as many as 1,024 offspring, which burst out as free-swimming parasites that must find a fish within about a day. Only that free-swimming stage can be killed, so a treatment has to keep the water lethal long enough to catch every wave. At 75 to 79°F (24 to 26°C) the cycle runs 3 to 6 days, and in cold water it stretches to weeks.

FOR A GOLDFISH	WHAT TO DO
Heat	The 86°F (30°C) that clears ich in tropical fish is far above a goldfish's ceiling. Warm only as far as 75°F (24°C), with an air stone, to speed the cycle.
Salt	About one tablespoon of aquarium salt per 5 gallons, added with time to acclimate. A goldfish is scaled, so salt suits it.
Medication	Malachite green at full strength, 5 ml per 10 gallons every 24 hours with a one-third water change before each dose. Never halve the dose. No improvement in 5 days means the diagnosis was probably wrong.
How long	Until every spot has been gone for days and no fish has died. The spots vanishing only means the parasites left to encyst. Expect a longer course than in a tropical tank.

- Treat the whole tank the fish are in, since the cysts sit on the gravel. Vacuum the gravel at each change, and take out snails and shrimp first.
- Pull the carbon and run an air stone through the course.
- Quarantine infected fish for at least 30 days before introducing any new fish to their tank.

Fin rot, fungus, ulcers, popeye & dropsy

Conditions that look unrelated and share one underlying cause: something has weakened the fish.

Fin rot

Frayed or disintegrating fin edges, usually bacterial and tied to stress and water quality. Test the water and correct it first, and keep up the water changes. There is no home antibiotic dose to reach for: antibiotics are rarely used for fin rot, because the drug struggles to reach the fin tips and medication in the tank harms the filter bacteria. Clean water, a good diet and a stable tank let the fish clear most cases, and the fins regrow. A case that keeps spreading past clean water tests is a vet case, where treatment can mean antibiotic injections and cleaning or trimming the infected tissue.

Fungal infections

Gray-white, cottony patches that develop after an injury or a period of stress. They typically appear on damaged skin or a healing wound, not on intact skin. Keep handling to a minimum and decor smooth, and find the injury or stressor that let it in.

Ulcers

An open sore, sometimes shallow, sometimes deep enough to expose muscle. They are typically bacterial, with *Aeromonas* a common culprit, tied to poor water, organic buildup, handling stress or an injury that gave the bacteria an opening. Advanced lesions can become deep, crater-like abscesses. This is not a shelf-medication problem: treatment is antimicrobial therapy based on sensitivity testing, which means a vet identifies which antibiotic will actually work.

Popeye

Swelling that pushes one or both eyes outward. One eye points more often to an injury or a local infection, both eyes more often to something systemic, commonly kidney or gill dysfunction that lets fluid build up behind the eyes and is frequently traced to water quality. Trauma is the most common cause overall. Some fancy varieties, telescope eye especially, are bred with naturally prominent eyes, so compare with that fish's normal look. Test the water. If it is clean and there is no obvious injury, it is a vet case.

Dropsy

A swollen belly with scales standing out in a pinecone pattern, best seen from above, or severe bloating. It is a visible sign of fluid building up in the body, usually from organ or kidney failure, and it frequently carries a poor prognosis. There is no safe home dose: treatment depends on the cause and usually combines antibiotics, corrected water and raised salinity, with the salt level set by a vet for that fish. Over-the-counter medication can harm the tank's bacteria and make the fish worse. Some fish caught early improve within days to weeks.

PINECONE SCALES ARE AN EMERGENCY

Get a fish vet promptly. A goldfish that has stopped eating or can no longer swim properly is the point where humane euthanasia becomes the kinder choice (page 32).

Flukes, anchor worm & velvet

Three parasite problems that do not look like the spots of ich, and are easy to miss early.

Gill and skin flukes

Flatworms (*Dactylogyrus* and *Gyrodactylus*) roughly 0.1 to 0.8 mm long, too small to see on the fish. You see the reaction instead: flashing against decor and substrate, gasping or rapid breathing at the surface, and gill covers held open over gills that look swollen and pale. Confirming flukes takes a skin or gill scrape examined under a microscope, not a look at the fish. Praziquantel is the standard treatment, as a prolonged bath at 5 mg/L, and it belongs in a vet's hands.

Anchor worm

Lernaea is a thin, thread-like parasite anchored in the flesh, with a red, inflamed, sometimes ulcerated point where it burrows in to feed. It is especially common in wild-caught and pond-raised fish, so a new pond addition or a fish from a crowded holding tank deserves a closer look before it joins your stock. Pulling the thread out with forceps is often incomplete and does not touch the parasite's other life stages, so more tend to appear later. The treatment is diflubenzuron at 0.066 mg/L, which is classed as a restricted-use pesticide in some regulatory settings. Source it and apply it on the guidance of a vet or an aquatic animal health professional.

Velvet

Piscinoodinium shows as a fine gold or rust-colored dust across the body, and it is harder to catch than ich. Regular tank lighting washes it out, so angle a flashlight across the fish from the side and the dust shows as a fine sheen. It is not cosmetic: infections in freshwater fish can cause high mortality once established.

FLASHING WITH NO SPOTS

Unexplained flashing, gasping or irritation with no visible spots is worth checking with a scrape under magnification, or a flashlight held at an angle, before you assume it is nothing. Test the water first, since a water problem looks the same.

WHERE TO TREAT

Treat in a hospital tank, not the display, except for ich, which lives in the whole tank. A new fish quarantined for 30 days and a clean water record prevent most of what is on this page.

Swim bladder disorder & constipation

Less dramatic than the last pages, and far more common in fancy varieties.

What it looks like

Buoyancy trouble: floating at the surface, sinking to the bottom, or swimming at a persistent angle. It is much more common in round-bodied fancy varieties, whose swim bladder is connected directly to the digestive tract. Air swallowed at the surface can end up in it, and gas buildup from excess or low-fiber food presses on it.

The usual triggers

- Overfeeding.
- Constipation.
- Gulping air at the surface while taking floating food.

The first response

1. Measure portions, and switch to sinking pellets instead of floating flakes.
2. A short fast.
3. Then a single cooked, de-shelled pea as a short-term aid.

Diet is always the first thing to check, but not every case clears with peas alone. Abnormal buoyancy that persists points toward something internal: a swim bladder problem, bloating, or something systemic. Persistent loss of buoyancy is a reason to see a fish vet.

WHY A PEA WORKS

The fix is almost comically simple because swim bladder problems are so often tied to constipation and diet in a round-bodied fish rather than to anything more serious. Peas are a short-term aid and should not become a staple.

FEEDING TO PREVENT IT

Sinking pellets, gel food, measured portions and no extra meals are the prevention. Adults are fine going 3 to 4 days without food.

Reading waste & when euthanasia is kindest

Reading waste

Once the water checks out, waste is one of the few internal health signals you can see without a microscope or a vet visit. Normal fish waste is a solid, somewhat compact strand that roughly matches the color of the food and does not linger behind the fish for more than a few seconds. The signal to watch is a shift toward pale, clear or white waste trailing in a long, stringy thread.

IT LOOKS LIKE	USUALLY MEANS	DO
A compact strand that matches the food and drops away	Normal	Nothing
Pale, stringy, trailing, a single episode in an eating, active fish	Nonspecific	Note it and recheck. Do not panic and do not treat
The same over several days, with weight loss or a pinched, sunken belly	A pattern that fits internal worms, or a fish that has stopped eating	Isolate it in a hospital tank and call a vet

It is nonspecific because the same sign fits several causes. Weight loss with a pot-bellied look fits internal worms, weight loss with refusing food fits an intestinal parasite, and a fish that simply stopped eating for any reason, including bad water, often passes mostly mucus that looks identical.

When euthanasia is the kindest option

Some fish will not recover, and a fish vet is the one to judge whether the case is past treatment. The first choice is a vet who treats fish: an overdose of anesthetic is the humane method. Where no vet can be reached, the method for aquarium fish is a clove oil overdose, since clove oil contains the sedative eugenol.

1. Use about 0.4 ml of clove oil per liter of water, in a container of water taken from the fish's own tank.
2. Mix the oil with a little warm water first, and add it slowly over about five minutes.
3. The fish loses consciousness, stops breathing and dies. Leave it in the solution for at least 30 minutes after the last gill movement.
4. Only then, as a final check, can it go in the freezer until frozen.

METHODS TO RULE OUT

Flushing puts a live fish into chlorinated water. Freezing a live fish, boiling, cutting off the head without first stunning the fish, and leaving it out of water to suffocate all cause the suffering the exercise is meant to end.

Setup checklist & targets

Essential setup checklist

- ☐ 20 gal (76 L) per fancy plus 10 gal (38 L) each extra, 40 gal (151 L) for a shubunkin, 55 gal (208 L) for a common or comet
- ☐ Filtration rated for 10 times the tank volume an hour or more
- ☐ Air pump and sponge filter as extra filtration and outage insurance
- ☐ Liquid reagent kit: ammonia, nitrite, nitrate, pH
- ☐ Dechlorinator labeled for chlorine and chloramine
- ☐ Bare bottom, fine sand, or gravel at least 1/2 in (1.3 cm) across
- ☐ Gravel vacuum and two buckets used only for the tank
- ☐ Soft rubber net and a transfer container
- ☐ Thermometer and a tight lid or mesh top
- ☐ Sinking goldfish pellets. A heater only if the room falls below 65°F

Water targets

READING	TARGET	ACT WHEN
Ammonia	0 ppm	Above 0
Nitrite	0 ppm	Above 0
Nitrate	Under 20 ppm	Over 40 ppm
pH	6.5 to 7.5	Moving fast. 6.5 to 9.0 is the reference range
Temperature	65 to 75°F (18 to 24°C)	Changing faster than 1°F (0.5°C) an hour
KH	Above 100 mg/L	At or below 50 mg/L
Dissolved oxygen	Above 5 mg/L	2 to 4 mg/L, with gasping at the surface
Water change	25 to 50% weekly	Nitrate past 20 ppm between changes

Before your goldfish comes home

- ☐ Tank fully cycled: a full ammonia dose reads zero within 24 hours, nitrite zero, nitrate present. Four to six weeks
- ☐ Filter and air pump running, outflow breaking the surface
- ☐ Bare quarantine tank set up with a seeded sponge filter and air pump
- ☐ Fish vet found and the number written on the emergency card

Emergency & quick targets

TEST THE WATER, THEN CALL A FISH VET, IF YOU SEE

Scales standing out in a pinecone pattern. Gasping at the surface with laboring gills. Open sores or red inflamed patches. A protruding eye. Persistent floating, sinking or swimming at an angle. Fin erosion advancing day by day. Refusing food for 24 to 48 hours alongside any other sign. More than one fish sick at once.

READING	TARGET	EMERGENCY ACTION
Ammonia	0 ppm	At 0.1 ppm change at least a third. Clearly climbing: 50% or more now, repeat
Nitrite	0 ppm	The same
Nitrate	Under 20 ppm	Over 40 ppm: a 50% change now, retest in two days
pH	6.5 to 7.5	Do not chase it with bottled adjusters
Temperature	65 to 75°F (18 to 24°C)	Cool or warm no faster than 1°F (0.5°C) an hour
Gasping at the surface	Oxygen above 5 mg/L	Add aeration and surface movement, lights off, test

Your numbers

Fish vet _____

Tank volume and date set up _____

Vet phone _____

Filter model _____

Local aquarium shop _____

Media last rinsed _____

HAVE THIS READY WHEN YOU CALL

Your recent test readings with dates, the tank volume and filtration, how many fish are in it, how long the signs have been going on, and everything you have already added to the water.

EMERGENCY WATER CHANGE, THE SHORT VERSION

Fill a clean bucket, add dechlorinator for the whole bucket, check the temperature is within a couple of degrees of the tank, siphon the old water from the substrate up, then refill slowly. Do not remove the fish. Do not deep-clean the filter on the same day.

Budget & shopping list

United States retail ranges, checked September 2026. Treat them as a snapshot, not a quote.

INITIAL SETUP	TYPICAL COST
Tank, 20-gallon minimum for one fancy goldfish	\$50 to \$150
Basic filtration at \$5 to \$15 per 10 gal of tank, so two units for 20 gal	\$10 to \$30
Heater, optional, mainly to stabilize temperature	\$15 to \$30
Substrate, dechlorinator, liquid test kit and decor, combined	\$30 to \$60
Equipment total	\$105 to \$270
The fish itself, more for fancy varieties	Often under \$10
Equipment plus the fish	About \$105 to \$280

MONTHLY ONGOING	TYPICAL COST
Pellets, occasional gel or frozen food, and periodic water treatments	\$10 to \$30

Shopping list

- ☐ Tank of 20 gal or more (55 gal or a pond for a common or comet)
- ☐ Canister or strong hang-on-back filter, rated well above the tank's size
- ☐ Liquid test kit and a water conditioner
- ☐ Gravel vacuum and two dedicated buckets
- ☐ Air pump, airline and sponge filter
- ☐ A bare bottom, fine sand, or gravel at least 1/2 in across
- ☐ Plants, hardscape and smooth decor, nothing sharp or fin-trapping
- ☐ Soft rubber net, transfer container, thermometer and lid
- ☐ Sinking goldfish pellets
- ☐ A spare tank for quarantine

NOT IN THE TOTALS

A canister filter costs more than the basic figure counted above. The spare quarantine tank is part of the real setup cost, and it is not priced here. A fish vet is a specialist service and is priced like one.

WHERE NOT TO CUT CORNERS

The tank and the filter. The extra \$50 to \$100 spent upfront on a proper tank and filtration is the difference between a fish that struggles and one that can live well over a decade.

First 30 days

Before day 1

- ☐ Tank cycled and confirmed: a full ammonia dose reads zero within 24 hours, nitrite zero, nitrate present
- ☐ Quarantine tank ready: bare, with a sponge filter seeded from a mature filter, and an air pump
- ☐ Fish vet found and written on the emergency card (34)

Day 1

- ☐ Float the sealed bag 20 to 30 minutes, then net the fish across and discard the bag water
- ☐ Match the temperature first; go slower if the bag and tank differ by more than 10°F (6°C)
- ☐ The newcomer goes into quarantine, not the display tank

Week 1

- ☐ Check daily for clamped fins, spots, labored gills, flashing, appetite loss and cloudy eyes
- ☐ Feed normally, once a day for an adult, and pull out whatever is left after a couple of minutes
- ☐ Test ammonia and nitrite in the quarantine tank, whose filter is small and new

Weeks 2 to 3

- ☐ Weekly 25 to 50% change in the quarantine tank and the display, testing first
- ☐ Start the Owner Log with the weekly readings (41)
- ☐ Rinse mechanical media in removed tank water only

Week 4 and beyond

- ☐ Day 30 is the quarantine floor, not the target: a fish you especially do not want to lose stays 30 to 60 days
- ☐ Review the log. Nitrate under 20 ppm before each change means the schedule fits; a drifting nitrate means feed less, change more, or stock fewer fish
- ☐ Before the fish joins an established tank, confirm it has shown no new signs and the quarantine tank is clear

Symptom quick reference

Test the water before acting on any row. It is the answer more often than anything else here.

SYMPTOM	LIKELY CAUSE	ACTION
Gasping at the surface, gills laboring	Low oxygen, or ammonia and nitrite burning the gills	Test now, water change, add surface movement (28)
White spots like grains of salt	Ich	Treat the whole tank until spots are gone for days (28)
Fine gold or rust dust, seen with a flashlight	Velvet	Vet; it can kill (30)
Flashing and rapid gilling, nothing visible	Flukes, or a water problem	Test first, then a vet scrape (30)
Thread-like projection with a red point	Anchor worm	Vet. Do not pull it out (30)
Frayed or receding fins	Fin rot	Clean water; a spreading case is a vet case (29)
Gray-white cottony patches	Fungus on an injury or stress point	Find the injury or stressor (29)
Scales raised in a pinecone pattern	Dropsy, usually organ or kidney failure	Fish vet promptly (29)
Open sores or red patches	Bacterial ulcer	Vet, water first (29)
One or both eyes protruding	Injury if one, systemic if both	Test water, then vet (29)
Floating, sinking or swimming at an angle	Swim bladder, usually diet	Sinking food, a short fast, then a pea (31)
Pale, stringy trailing waste	Nonspecific	Watch appetite and weight (32)
Same, with weight loss over days	Internal worms or a fish not eating	Hospital tank and a vet (32)
White pimples on the gill covers and pectoral fins	Breeding tubercles on a ready male	Nothing (24)
Chasing in late spring	Spawning behavior	Normal (24)
Off food for 24 to 48 hours plus another sign	Several causes	Test water, then a fish vet (18)
Any change with no obvious cause	Several possible causes	Fish vet, with the Owner Log (41)

Daily, weekly & seasonal routine

Daily

- ☐ Feed the measured portion and pull out what is left after a couple of minutes
- ☐ Check the filter is running and the surface is moving
- ☐ Look at every fish: fins open, breathing steady, no spots, all accounted for

Weekly

- ☐ Test ammonia, nitrite, nitrate and pH on the same day, and log them
- ☐ Change 25 to 50 percent of the water, siphoning the substrate
- ☐ Check the temperature

Monthly

- ☐ Rinse mechanical media in removed tank water, never under the tap
- ☐ Replace carbon if you use it, and only part of the biological media at a time
- ☐ Review the log for slow trends, not just single readings

Seasonal

- ☐ Summer: watch for the water climbing past 75°F (24°C) and add surface agitation
- ☐ Pond only: scale feeding down with temperature and stop below 39°F (4°C)
- ☐ Spring: expect chasing and a dip in appetite in a mature mixed group

NITRATE SETS THE SIZE OF THE CHANGE

Under 20 ppm before the change, keep the schedule. 20 to 40 ppm, change 40 to 50 percent weekly or add a midweek change, and feed less. Over 40 ppm, change 50 percent now and retest in two days.

CHECK EVERY FISH, EVERY DAY

A change in behavior shows up before a mark does. A fish that is hiding, isolating or resting somewhere unusual is worth a closer look the same day.

Power outages, heat waves, travel & transport

For a fish, a power outage is an oxygen problem first and a temperature problem second.

When the pumps stop, the surface stops moving and gas exchange stops with it. Dissolved oxygen starts falling within a couple of hours, sooner in a heavily stocked or warm tank. A goldfish tolerates cold better than a tropical fish, but goldfish are often kept at high stocking densities, so the oxygen math matters just as much.

Do

- Aerate right away. A battery-powered air pump and air stone is the cleanest fix.
- With no pump, scoop water out with a cup or pitcher and pour it back from a few inches up for a minute or two. Repeat roughly every 10 to 15 minutes in a small tank and every 20 to 30 minutes in a larger one.
- Unplug the filter deliberately, so it does not restart on its own.

Don't

- Don't feed until the filter is back and stable. Uneaten food consumes oxygen.
- Don't restart a filter that sat off for hours without rinsing the media in old tank water or at least looking inside.
- Don't add a space heater, heat lamp or open flame near the tank.

A HEAT WAVE

Heat is the other oxygen problem, and the ranked steps are on page 25.

Leaving town

TIME AWAY	WHAT IS NEEDED
Up to 3 to 4 days	Nothing special for a healthy adult in an established tank
Beyond about 5 days	Effects on health may start to show. Line up a sitter or a feeder
A week to 10 days	Arrange a feeder. Use the Pet-Sitter Sheet (40)

Transport and vet trips

- A bag beats a rigid bucket for most short trips: it holds the fish in the dark, which is calming, and flexes instead of sloshing. Use a tub or bucket with a tight lid for a longer trip.
- Leave real headspace above the water. For a long trip, a battery pump or pure oxygen is standard. Keep it out of direct sun.
- Move the fish last, once the new tank's water is dechlorinated and close to the old temperature, and get it out of the bag soon.
- Pack the filter media in a sealed bag or tub of the tank's own water, never tap water, so the colony survives.

FILL IN AND LEAVE BY THE TANK

Pet-sitter sheet

Everything a sitter needs on one page. Walk them through it once in person before the first solo visit.

Tank volume and number of fish _____

Fish vet _____

Fish names or descriptions _____

Local aquarium shop _____

My phone _____

Backup contact _____

Each visit

- ☐ Check the filter is running and the water surface is moving
- ☐ Count the fish and check each one is swimming normally
- ☐ Feed only the portion I left out, and net out anything uneaten after a couple of minutes
- ☐ Check the temperature reads between 65 and 75°F (18 and 24°C)
- ☐ Wash your hands afterward

PLEASE DON'T

Feed more than the portion I left, change, clean or unplug the filter, top up the tank with untreated tap water (it may hold chloramine), or add anything at all to the water.

CALL ME IF

The filter is silent or the surface is still. A fish is gasping at the surface, lying on the bottom, floating oddly or covered in spots. Any fish has died. The power has been out for more than a few hours. If I don't answer, call the vet number above.

IF A FISH DIES WHILE I'M AWAY

Net it out and leave the rest of the tank alone. Do not do a big water change or clean the filter to compensate. Take a photo, message me, and check the other fish at the next visit.

Owner log

One row a week, filled in on test day. Photocopy this page, or track digitally.

DATE	AMMONIA	NITRITE	NITRATE	PH	TEMP	% CHANGED	APPETITE, BEHAVIOR, NOTES

WHY BOTHER LOGGING

For a fish, the water readings are the vital signs, and they take two minutes and no handling. Most problems announce themselves in the numbers before they show on the fish. Sixteen rows is four months, enough to see a trend rather than a bad day.

Equipment & vet log

Equipment log

ITEM	INSTALLED	REPLACE OR CHECK BY	DONE
Tank, volume and date set up			
Filter, model and rated flow			
Biological media, portion last replaced			
Air pump, airline and air stone			
Thermometer			

Twelve-month planner

Tick the box in the month you did it. Start with whichever month you are in.

TASK	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Rinse mechanical media	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Replace carbon, if used	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Part of the biological media	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Quarantine tank dry and ready	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Vet visits

DATE	VET	REASON AND WHAT WAS DONE	FOLLOW-UP

Incident & treatment log

Record every medication, salt dose and water emergency. Doses accumulate, and a vet will ask.

DATE	PROBLEM AND READINGS	WHAT WAS ADDED, AND HOW MUCH	OUTCOME

Enrichment checklist & log

Setup

- ☐ An open swim lane running the length of the tank
- ☐ Planted or structured cover at both ends, live or artificial
- ☐ Sand, or gravel at least 1/2 in across, for foraging
- ☐ Sinking food as the default

Monthly review

- ☐ Check nothing has developed a sharp edge or a fin trap
- ☐ Check hardscape is still secured
- ☐ Try a safe vegetable from the foods page (16)

Enrichment log

DATE	ACTIVITY	HOW IT WENT, AND WHETHER TO REPEAT

ENRICHMENT IS NOT EXTRA FOOD

Every foraging idea here uses the fish's existing daily ration, divided differently. A second meal on top of it is overfeeding, not enrichment.

Glossary, A to H

Air stone	A porous block on an air pump's tube that releases fine bubbles. It moves water and breaks the surface, which adds oxygen. Page 9.
Ammonia	The first and most toxic waste product, from fish waste and decaying food. The target is 0 ppm in a cycled tank. Page 11.
Ammonia-binding conditioner	A water conditioner that binds or neutralizes ammonia, buying a margin of safety between water changes while the bacteria catch up. Page 10.
Anchor worm	A thread-like parasite (<i>Lernaea</i>) anchored in the skin, with a red, inflamed point where it burrows in. Page 30.
Bioload	The total waste a tank's filter and water volume have to process. A goldfish's is unusually high for its size. Page 8.
Biological media	Ceramic rings or sponge in a filter that hold the bacteria that turn ammonia into nitrite and nitrite into nitrate. Rinse gently in old tank water, never under the tap. Page 9.
Blanching	Putting a vegetable in boiling water for a minute or two, then cooling it, so a goldfish can break it apart and eat it. Page 16.
Chloramine	Chlorine bonded to ammonia, used in many water supplies. Unlike chlorine it does not gas off when water stands, so a conditioner is always needed. Page 11.
Clamped fins	Fins held tight against the body instead of spread. One of the more universal signs of stress or illness. Page 20.
Cycling	Growing the bacteria that convert ammonia to nitrite and nitrite to nitrate. It takes roughly four to six weeks in a new tank. Page 10.
Dechlorinator	A water conditioner that removes chlorine, and for chloramine a product labeled for both, from tap water before it reaches the fish. Page 12.
Diflubenzuron	A medicine that kills anchor worm, used at 0.066 mg/L and only on a vet's guidance. Page 30.
Dissolved oxygen	The oxygen in the water, in mg/L. Above 5 supports optimum health, 2 to 4 is distress, and below 2 is deadly. Page 7.
Dropsy	Not a disease but a sign: fluid swelling that lifts the scales into a pinecone pattern, usually from organ or kidney failure. Page 29.
Fancy goldfish	The round-bodied ornamental varieties, such as oranda and ranchu. Slower swimmers, most prone to buoyancy trouble. Page 5.
Fin rot	Frayed or disintegrating fin edges, usually bacterial and tied to poor water. Page 29.
Flashing	Rubbing or scraping the body against surfaces, a sign of irritation from parasites or poor water. Page 20.
Flukes	Microscopic flatworms on the gills and skin. They cause flashing and fast breathing with nothing visible. Page 30.
GH	General hardness: the dissolved calcium and magnesium in the water. Moderately hard to hard water (61 to 180 mg/L) suits goldfish. Page 11.
gph	Gallons per hour: how much water a filter pumps. Ten times the tank volume an hour or more is the goldfish target. Page 8.
Hospital tank	A bare, separately cycled tank for quarantine and treatment, so medication does not reach the display tank's bacteria. Page 23.

Glossary, I to Z

Ich	White spot disease, caused by <i>Ichthyophthirius multifiliis</i> . Only its free-swimming stage can be killed, so treatment must outlast the visible spots. Page 28.
Infusoria	Tiny living creatures, the first food for free-swimming fry. Page 14.
KH	Carbonate hardness: the water's buffer against pH change. Above 100 mg/L is recommended, and at or below 50 mg/L the filter bacteria suffer. Page 11.
Malachite green	A dye-based medicine used against ich, at full strength, 5 ml per 10 gallons, never halved. Page 28.
mg/L	Milligrams per liter, a measure of how much of a substance is dissolved in water. For most water tests it reads the same as ppm. Page 11.
Nauplii	Newly hatched baby brine shrimp, the standard first food for free-swimming fry. Page 14.
Nitrate	The end product of the nitrogen cycle. Far less toxic than ammonia or nitrite, but kept under 20 ppm with water changes. Page 12.
Nitrite	The middle product of the nitrogen cycle, also toxic. The target is 0 ppm. Page 11.
Nitrogen cycle	The chain in which bacteria turn ammonia into nitrite and nitrite into nitrate. Page 10.
NYCRR	The New York Codes, Rules and Regulations, the state's rulebook. Part 575 lists the regulated invasive species that include the goldfish. Page 26.
pH	How acidic or alkaline the water is. A goldfish does best at 6.5 to 7.5, and a steady reading matters more than the number. Page 11.
Piping	Gulping air at the surface. It is usually low oxygen, not infection. Page 20.
Popeye	One or both eyes pushed outward, from injury, infection or fluid buildup behind the eye. Page 29.
ppm	Parts per million, a measure of how much of a substance is dissolved in water. For most water tests it reads the same as mg/L. Page 11.
Praziquantel	A medicine for flukes, given as a prolonged bath at 5 mg/L under a vet's guidance. Page 30.
Quarantine	Keeping a new or sick fish in a separate tank for at least 30 days before it meets other fish. Page 23.
Single-tail	The streamlined varieties: common, comet and shubunkin. Larger and faster, and much better suited to a pond than a tank. Page 5.
Slime coat	The living protective layer over a goldfish's skin. It takes 24 to 48 hours to regenerate if handling strips it. Page 19.
Stunting	Suppressed outward growth from cramped space and chronic ammonia while the organs keep developing. A welfare problem, not a size feature. Page 5.
Swim bladder	The internal organ that controls buoyancy. Problems with it cause floating, sinking or swimming at an angle. Page 31.
Tubercles	Small white pimples on the gill covers and pectoral fins of a male ready to spawn. Not disease. Page 24.
Turnover	How many times an hour a filter processes the tank's total volume. Page 8.
Ulcer	An open sore in the skin, usually bacterial, needing a vet. Page 29.
Units	°F and °C: degrees Fahrenheit and Celsius. in, cm, mm, ft: inches, centimeters, millimeters, feet. gal and L: gallons and liters. ml: milliliters. lb: pounds.
Velvet	A fine gold or rust-colored dust on the skin from the parasite <i>Piscinoodinium</i> , best seen with a flashlight. Page 30.

Sources

Where the numbers in this package come from, so you can check them yourself.

Veterinary references

- Merck Veterinary Manual: management of aquarium fish; routine health care of fish; overview of aquarium fish; parasitic, bacterial and environmental diseases of fish; disorders and diseases of fish; reference ranges for water quality; providing a home for fish; aquatic systems. The 30-day quarantine, the 6.5 to 9.0 pH range, the 1°F an hour limit, the fluke, ulcer and ich guidance.
- University of Florida Institute of Food and Agricultural Sciences Extension: dissolved oxygen for fish production; ich; anchor worm; ammonia in aquatic systems; fish nutrition and preparing fish feeds; ornamental fish cultured in Florida; koi herpesvirus; on-farm transport. The oxygen figures, the ich life cycle, the anchor worm dose, food storage and transport chemistry.
- Aquatic Veterinary Services and The Fish Vet (Jessie Sanders and Richmond Loh): fin rot, popeye, nitrate, old tank syndrome, dechlorinators, koi versus goldfish, and whether goldfish have a stomach. PetMD, veterinarian-reviewed: the goldfish care sheet, swim bladder disorders and dropsy.
- American Association of Fish Veterinarians: the Find a Fish Vet directory and its three fallback steps. American Veterinary Medical Association: selecting a pet fish. VCA Animal Hospitals: fish white spot disease.
- Royal Society for the Prevention of Cruelty to Animals, Australia: the most humane way to euthanize aquarium fish. North Carolina State University College of Veterinary Medicine: aquariums and ponds during power outages.

Research

- Ford and Beitinger (2005), Journal of Thermal Biology 30: temperature tolerance in the goldfish. Herrera-Castillo and colleagues (2024), Biology: goldfish metabolic rate and warming.
- Smith and Gray (2011), Animal Welfare 20: substrate and goldfish foraging. Sullivan and colleagues on goldfish habitat preference, summarized by Companion Animal Psychology. Sibeaux and colleagues (2022), Proceedings of the Royal Society B: goldfish estimate distances.
- Smith and colleagues (2012), PLOS One: microbial diversity and pathogens in ornamental fish aquarium water. Gomelsky, Schneider and Plouffe (2012), North American Journal of Aquaculture 74: koi and goldfish hybrids.
- Gomez Isaza, Cramp and Franklin (2020), Conservation Physiology: nitrate and low pH in a freshwater fish. The 2025 ISME Communications study of home aquarium filters. United States Geological Survey: pH, hardness and alkalinity. Southern Regional Aquaculture Center: starting a biofilter.

Public health, government and records

- Center for Food Security and Public Health, Iowa State University: *Mycobacterium marinum*. Minnesota Department of Natural Resources: goldfish as a regulated invasive species. New York State, Codes, Rules and Regulations, Part 575. Jamaica Ministry of Agriculture, Fisheries Division: breeding goldfish. Guinness World Records: the oldest goldfish, Tish.

Husbandry references

- Aquarium Co-Op: ich, goldfish plants, stocking and water changes. William H. Wildgoose, veterinarian, and EWASH: gravel in goldfish mouths and handling. The Goldfish Tank: slime coat and handling. Practical Fishkeeping: cycling and ammonia. Advanced Aquarium Concepts, The Vet Desk and Aquarium Buddies: goldfish care and price checks.

REFERENCE

Where the sources disagree

Where credible sources give different numbers, this package prints one and says why.

TOPIC	WHAT THE SOURCES SAY	WHAT THIS PACKAGE USES
Filter turnover	No governing body publishes one figure. Sizing charts give about 4 times an hour for a lightly stocked tank, 6 to 8 for a community, and 10 or more for a heavily stocked or messy one	10 times an hour or more, since a goldfish tank is the messy case
Heater	A veterinarian-reviewed care sheet recommends a thermostatic heater; a goldfish is a coldwater fish comfortable at 65 to 75°F and can be stressed by a heater running too warm	No heater unless the room falls below 65°F
pH	6.5 to 7.5 as a goldfish target; 6.5 to 9.0 as the general veterinary range for freshwater fish	Aim for 6.5 to 7.5, and treat 6.5 to 9.0 as workable if steady
Nitrate limit	Most fish tolerate about 20 mg/L; 40 is the ceiling below which most are considered safe	Under 20, and act past 40
Water change size	25 to 30 percent weekly as the general baseline; goldfish given 25 to 50 percent as the heaviest waste producer for its size	25 to 50 percent weekly, set by nitrate
Fish-in action line	0.1 ppm (a third of the tank); 0.1 to 0.25 ppm in day-to-day practice	Act at 0.1 ppm, and 50 percent or more once clearly climbing
Quarantine	At least 30 days; 30 to 60 days for a fish you especially do not want to lose	30 days as the floor, 30 to 60 for a valued fish
Anchor worm dose	Diflubenzuron at 0.066 mg/L for <i>Lernaea</i> ; a lower general figure of 0.03 mg/L for crustacean parasites also circulates	0.066 mg/L, the dose written for <i>Lernaea</i>
Going without food	3 to 4 days with no concern; effects may show past 5 days; survival cited at 1 to 2 weeks	3 to 4 days is fine; arrange a feeder past a week
Pond winter feeding	A pond goldfish slows into dormancy and eats little below roughly 50°F; feeding stops entirely below 39°F	Normal above 50°F, a wheat-germ formula from 39 to 50°F, none below 39°F

Version history & about

What changed in each edition, so a printed copy can be placed.

Version history

This is edition 3.0 (Oct 2026). The store page shows the current edition; re-download free if yours is older. Where this list and a page disagree, the page wins.

- **3.0, Oct 2026.** Rebuilt on the current outline, 48 pages. Safety: pH 6.5 to 7.5; fishless cycle dosed to 3 ppm; fish-in change of at least a third at 0.1 ppm; bag floated 20 to 30 minutes; adults fed once a day, or the same ration split in two, with no weekly fasting day; ich treated with salt or malachite green until the spots have been gone for days, heat no higher than 75°F; anchor worm dose 0.066 mg/L; no home antibiotic for fin rot; floor bare, fine sand or gravel at least 1/2 in across; temperature moved no faster than 1°F an hour; outage surface agitation every 10 to 15 minutes in a small tank, 20 to 30 in a larger one; goldfish and koi can be crossed; never-feed list is bread and refined starches, raw meat, cheese and onions. Legal: new page on never releasing a goldfish, with the Minnesota and New York rules. Costs: equipment \$105 to \$270, monthly \$10 to \$30. Other changes are wording only.
- **2.1, Sep 2026.** Safety: turnover raised to 10 times an hour; KH retargeted to 100 ppm and up; GH 60 to 180 ppm; quarantine raised to a 30-day minimum; fish-in action line 0.1 to 0.25 ppm. Replaced by 3.0.
- **2.0, Sep 2026.** Rebuilt and expanded from 21 to 39 pages, with a fish-in cycling plan, a power-outage plan and a pet-sitter sheet. The owner log began recording water readings instead of weight. Replaced by 2.1.
- **1.0, Aug 2026.** First edition, 21 pages. Replaced by 2.0.

This package is a husbandry reference compiled from the listed sources and cross-checked against current veterinary guidance at the time of publication. It is not a substitute for a veterinarian who treats fish, and nothing in it should delay a vet visit. Medication doses quoted here are from published references and are included so you can discuss them with a vet, not so you can dose a tank unsupervised. The law page is a summary as of October 2026, not legal advice.

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